

CA-BSNet Proposed System

Complete Proposed System — Final Locked Plan

Master's Thesis Proposal

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1. Base Paper Identification

Title: A Boundary-Semantic Synergy Learning Network for Breast Ultrasound Lesion Segmentation

Authors: Hao Yang, Lingxiao Jin, Yuqin Feng, Shuxian Liu — School of Computer Science and Technology, Xinjiang University

Venue: *Scientific Reports* (2026), DOI: 10.1038/s41598-026-67386-0

Model name: D-SAS (Dual-branch Semantic-Aware Segmentation Network)

Reported results: Dice 82.10% / IoU 70.21% on BUSI; Dice 89.14% / IoU 81.34% on BUSI_WHU; Dice 80.95% / IoU 72.82% on cross-dataset BUSI_WHU→UDIAT — outperforming 17 competing methods (U-Net, UNet++, TransUNet, SwinUnet, DeepLabV3+, and others) under 5-fold cross-validation with paired statistical significance testing.

Architecture: ResNet101 backbone → **SSBG** (Speckle Suppression Boundary Gate, operates on 1/4-resolution shallow features) + **AAS** (Adaptive Atrous Selector, four dilation rates 1/2/4/6 + global pooling branch, operates on 1/16-resolution deep features) → **SFH** (Synergistic Fusion Head, bidirectional gating + Softplus-weighted fusion) → progressive decoder.

This is a real, peer-reviewed, statistically validated state-of-the-art method, not a proposal document — it is the foundation this thesis builds on and the number this thesis’s own results must be measured against.

2. Decision Record — How This Plan Was Reached

Three candidate directions were evaluated before settling on the final plan:

Candidate	Verdict	Why
MAS-Net (AI-generated concept combining transformer bottleneck, dual attention, boundary branch, and full mean-teacher semi-supervised training)	Rejected	Never implemented or validated; depends on an unlabeled ultrasound dataset that was never identified; bundles four separate research problems into one system — too high-risk and too broad for a bounded Master’s contribution
Attention U-Net (Sulaiman et al., 2024, <i>Sci Rep</i>)	Adopted, but only as Tier 1	A real, simpler, published architecture — good as a pipeline sanity check before building D-SAS’s custom modules, but too modest a contribution to be the thesis’s core claim on its own
UGBR-Net (uncertainty-guided boundary refinement via Monte-Carlo Dropout, considered as an alternative modification to D-SAS)	Considered, not selected — kept as Future Work	A legitimate, well-scoped alternative modification targeting D-SAS’s own stated limitation (its fusion weights are “feature activation magnitudes rather than probabilistic uncertainty”). Ultimately not chosen over CA-BSNet because of higher inference cost (10-20x forward passes per image) and thinner reference material to build against for a first implementation on a compressed timeline
CA-BSNet (cross-attention block inserted between D-SAS’s AAS and SFH modules)	✔ SELECTED	Single, clearly bounded architectural change; cross-attention is one of the most standard, well-documented mechanisms in deep learning, giving abundant reference implementations to check work against; cheap to run (touches only the smallest feature map, single forward pass); directly targets a limitation the D-SAS authors state in their own introduction (that purely local convolutional fusion lacks global context, while full Transformer approaches are too computationally expensive for this task)

3. Proposed System: CA-BSNet

3.1 Proposed Title

Cross-Attention Enhanced Boundary-Semantic Synergy Network for Breast Ultrasound Lesion Segmentation (CA-BSNet)

3.2 Introduction

Breast ultrasound (BUS) is a widely used, low-cost screening modality, but accurate lesion segmentation remains difficult due to speckle noise, low contrast, and blurred lesion boundaries. D-SAS (Yang et al., 2026) achieves state-of-the-art results on this task through three complementary modules — SSBG for boundary enhancement, AAS for multi-scale semantic perception, and SFH for boundary-semantic fusion. This thesis builds directly on D-SAS and extends it in one specific, clearly-scoped direction: inserting a lightweight cross-attention mechanism between the AAS and SFH modules, testing whether a small amount of learned global relational modeling improves segmentation of large, irregular, or multifocal lesions — a limitation the original authors identify but do not address within their own architecture.

3.3 Problem Statement

Given a breast ultrasound image, the task is pixel-wise binary segmentation of lesion regions under conditions of speckle noise, blurred boundaries, and large variation in lesion scale and shape. This thesis specifically investigates whether AAS’s purely local, pixel-wise multi-scale gating under-represents relationships between distant image regions relative to a lightweight attention-augmented alternative.

3.4 Research Gap

Existing published work (D-SAS): provides boundary enhancement (SSBG), multi-scale local gating (AAS), and bidirectional gated fusion (SFH), demonstrating SOTA Dice/IoU/HD95 on BUSI and BUSI_WHU with strong cross-dataset generalization.

Proposed improvement: D-SAS’s own related-work discussion argues that full Transformer models are too computationally costly and that global context alone doesn’t resolve boundary/scale issues — but it does not test a *minimal*, single-block attention addition layered on top of its own fusion mechanism. This thesis fills that narrow, specific gap.

This work does not claim SSBG, AAS, or SFH as original contributions — they are reproduced from the published paper. The contribution is the cross-attention insertion, its ablation, and the resulting accuracy/efficiency analysis.

3.5 Objectives

1. Reproduce D-SAS (ResNet101 + SSBG + AAS + SFH) as a faithful baseline on BUSI.
2. Design and implement a lightweight cross-attention block operating on AAS output features, integrated ahead of SFH.
3. Train and evaluate CA-BSNet on BUSI under a consistent, documented protocol.
4. Run an ablation comparing D-SAS vs. CA-BSNet on Dice, IoU, HD95, with bootstrap-based significance testing.
5. Deliver a Streamlit demo application performing inference with the trained model.

3.6 Proposed Methodology

Input BUS image
→ Preprocessing (dedup, resize to 256×256, normalization, augmentation)

(Pawłowska, Karwat & Żolek) identified approximately 235 duplicate or near-duplicate images in BUSI, created by affine-transformed re-submissions of the same underlying cases. These must be removed before splitting the dataset, or duplicates leaking across train/test will inflate reported Dice/IoU in a way that will not replicate. This dedup pass is performed in Week 1 and documented in the methodology section.

Secondary (not required for core thesis): BUSI_WHU - 927 images, official train/val/test split — Mendeley Data: <https://data.mendeley.com/datasets/k6cpmwybk3/1> (also mirrored on Kaggle) - Freely downloadable, no request required - Optional external validation only; not on the critical path

Not used: UDIAT - Used by D-SAS for cross-dataset generalization testing, but **not freely downloadable** — requires emailing the original authors (Yap et al.) with no guaranteed response time. Excluded from the core plan to avoid a dependency outside the project’s control.

Training/validation/testing strategy: single stratified 70/15/15 split (train/val/test), applied identically across all three tiers of models for a fair, consistent comparison. This replaces 5-fold cross-validation to fit the compressed timeline; the trade-off is addressed via bootstrap resampling (Section 3.10).

3.9 Experimental Plan

Three tiers, evaluated under one identical protocol:

Tier	Model	Role
1	Attention U-Net (Sulaiman et al., 2024)	Pipeline sanity check — validates data loading, training loop, and metric code before custom modules are built
2	D-SAS (reproduced)	State-of-the-art baseline / comparison target
3	CA-BSNet	Proposed contribution

Training: identical hyperparameters and preprocessing across all three tiers so differences are attributable to architecture, not tuning.

Comparison methodology: bootstrap resampling (1000 iterations) on the fixed test set to estimate 95% confidence intervals on the metric differences between D-SAS and CA-BSNet — this substitutes for paired significance testing across folds, which a single-split protocol cannot otherwise provide.

3.10 Evaluation Metrics

- **Dice Score:** $2|P \cap Y| / (|P| + |Y|)$ — overlap between predicted and ground-truth masks.
- **IoU (Jaccard Index):** $|P \cap Y| / |P \cup Y|$ — a stricter overlap measure than Dice.
- **Precision:** $TP / (TP + FP)$ — of predicted lesion pixels, how many are correct.
- **Recall (Sensitivity):** $TP / (TP + FN)$ — of true lesion pixels, how many were found; the more clinically critical error to minimize.
- **Specificity:** $TN / (TN + FP)$ — of true background pixels, how many were correctly excluded.
- **HD95:** 95th-percentile boundary distance error — directly measures the boundary localization quality this thesis’s modification specifically targets.
- **Bootstrap 95% CI:** on the (CA-BSNet – D-SAS) difference for each metric above, from 1000 resamples of the test set.

3.11 Expected Contributions

- A working, reproducible re-implementation of a 2026 state-of-the-art breast ultrasound segmentation method.
- A clearly scoped, ablation-validated architectural addition (lightweight cross-attention), with quantified accuracy/efficiency trade-offs and bootstrap confidence intervals.
- A deployable Streamlit demonstration tool.

To be evaluated experimentally — no results are assumed or fabricated in this proposal.

3.12 Limitations

- BUSI is a relatively small dataset (~780 images, ~647 after excluding normal cases); results may not generalize to other ultrasound devices or populations without further cross-dataset testing.
- The original D-SAS trained weights are not available, so this reproduction may not exactly match the published numbers — normal re-implementation variance, to be disclosed rather than hidden.
- A single 70/15/15 split (rather than 5-fold CV) is used to fit the compressed timeline; bootstrap resampling partially compensates but does not fully replace fold-level cross-validation rigor.
- This is 2D slice-based semantic segmentation, not volumetric or instance segmentation — multifocal lesions are not separately counted.
- Cross-attention adds parameters and compute; if gains are marginal or absent, this will be reported as an honest negative/neutral result rather than overstated.

3.13 Future Work

- **Uncertainty-guided boundary refinement (UGBR-Net direction):** genuine model-derived uncertainty via Monte-Carlo Dropout, directly addressing D-SAS's own stated limitation that its SFH weighting maps represent feature activation magnitudes rather than probabilistic uncertainty. Deferred due to the added inference-time cost of multi-pass stochastic forward passes; a promising follow-up direction, particularly for its potential to produce clinically interpretable confidence maps.
- 5-fold cross-validation as a follow-up rigor pass, time permitting.
- Cross-dataset generalization testing (BUSI_WHU→UDIAT), if UDIAT access is obtained.
- Lightweight backbone replacement (EfficientNet-B0/MobileNetV3) for edge/point-of-care deployment, as suggested by the original D-SAS authors.
- Semi-supervised extension (mean-teacher), evaluated only if a genuine unlabeled ultrasound dataset becomes available.

4. User Interface Plan (Streamlit)

Inference only — the app loads a trained checkpoint; training remains a separate offline process.

1. Home — title, description, pipeline diagram, “Get Started” button.

2. Upload Ultrasound Image — file uploader (PNG/JPG), preview, “Run Segmentation” button.

3. Segmentation Result — original image, predicted mask overlay side by side; “Download Mask” button.

- 4. Performance Dashboard** — static bar chart comparing all three tiers (Attention U-Net, D-SAS, CA-BSNet) across Dice/IoU/HD95, with the bootstrap confidence interval shown for the CA-BSNet vs. D-SAS comparison. Read-only — no live recomputation.
- 5. About Project** — project background, dataset used, architecture summary, citation to the base D-SAS paper, clear statement distinguishing the reproduced components from the original contribution.

5. Final Locked Roadmap (8 Weeks)

Week	Focus
1	Download BUSI; dedupe (~235 duplicate/near-duplicate images per Pawłowska et al., 2023); create single stratified 70/15/15 split; build Dataset / DataLoader + augmentation pipeline
2	Build Attention U-Net (Tier 1); train; confirm Dice/IoU/SE/SP/HD95 metric code is correct
3	Implement ResNet101 backbone + SSBG module; unit-test each in isolation
4	Implement AAS + SFH modules; assemble full D-SAS (Tier 2); first end-to-end forward pass; begin training
5	Finish D-SAS training/tuning; generate test-set predictions; compare against Yang et al.’s published numbers (context only)
6	Implement the cross-attention block between AAS and SFH → CA-BSNet (Tier 3); train; generate predictions; run bootstrap resampling for the significance check
7	Build Streamlit app: Home → Upload → Segmentation Result → Performance Dashboard → About
8	Write-up: results table with confidence intervals, limitations, guide presentation document, viva rehearsal

6. Guide Presentation Summary

- Title:** Cross-Attention Enhanced Boundary-Semantic Synergy Network for Breast Ultrasound Lesion Segmentation (CA-BSNet)
- Background:** D-SAS (Yang et al., *Scientific Reports*, 2026) achieves state-of-the-art breast ultrasound lesion segmentation via three modules — SSBG, AAS, SFH — validated on the public BUSI and BUSI_WHU datasets.
- Problem:** D-SAS’s AAS module fuses multi-scale features via purely local, pixel-wise gating; the authors themselves note that full Transformer-based global context modeling is too computationally expensive for this task, but do not test a minimal attention addition on their own architecture.
- Research Gap:** No existing evaluation of whether a small, cheap, single-block cross-attention mechanism — added only at the smallest feature resolution, between AAS and SFH — improves D-SAS’s handling of large, irregular, or multifocal lesions without materially increasing cost.

Proposed Methodology: Reproduce D-SAS as a faithful baseline on BUSI (single stratified 70/15/15 split, given the compressed timeline). Insert a lightweight cross-attention block between AAS output and SFH input to form CA-BSNet, keeping every other module unchanged. Train both under identical hyperparameters; compare via Dice, IoU, SE, SP, HD95, with bootstrap-based confidence intervals substituting for fold-level significance testing.

Dataset: BUSI (780 public images with pixel-level masks, freely downloadable, deduplicated before splitting).

Expected Outcome: *To be evaluated experimentally.* Hypothesis: modest Dice/IoU improvement and/or better handling of large or irregular lesions, at some additional but small computational cost.

Technology Stack: Python, PyTorch, Albumentations, Google Colab, Streamlit, MONAI (optional).

Timeline: 8 weeks, per Section 5 above.

Final Recommendation

Selected Project: CA-BSNet — a cross-attention modification of D-SAS (Yang et al., 2026)

Reason: D-SAS is a real, peer-reviewed, reproducible baseline with public data and documented numbers, making a single well-scoped architectural change feasible on a compressed timeline and defensible in a viva.

Dataset: BUSI (primary, freely available, deduplicated); BUSI_WHU optional; UDIAT excluded (not freely available).

Proposed Modification: Lightweight multi-head cross-attention block on the AAS module's output, before SFH fusion.

Difficulty /10: 5 (medium — new technique for the author, but scoped to a single, small, well-understood module with abundant reference implementations).

Estimated Completion Risk: Low-to-Medium — main risks are re-implementation fidelity to the original paper and the attention block underperforming, either of which is still a valid, reportable result.

Why this is better than the alternatives: MAS-Net bundles too many unvalidated components with no identified unlabeled dataset for its core claim. UGBR-Net is a legitimate alternative but carries higher inference cost and thinner reference material for a first implementation — retained as a named Future Work direction rather than discarded.